

Experiment No.	1
Experiment Title	Working with Python Packages: NumPy, SciPy, Scikit-Learn, Matplotlib, and Pandas (Iris Botanical Dataset)
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1 Objective

The main goals of this exploratory analysis on the Iris dataset are:

- Leverage core Python numerical packages (NumPy, Pandas, Seaborn, Matplotlib) to perform automated statistical profiling on botanical morphometric features.
- Evaluate multi-dimensional linear separability between target species (Setosa, Versicolor, Virginica).
- Export diagnostic visual figures into `figures_iris/`.

2 Problem Statement

In supervised classification tasks, identifying attribute pairs that achieve clean hyperplanar separability significantly improves model training speed and accuracy. Manual feature plotting for multi-class targets can be tedious. This report uses an automated EDA pipeline to evaluate class overlap and feature correlation bounds across all four morphometric measurements of the Iris dataset.

3 Dataset Description

The **Iris Botanical Dataset** consists of 150 flower samples evenly divided across three species.

Dataset Name	Iris Botanical Dataset
Dataset Source	UCI Machine Learning Repository
Number of Samples	150 (50 per species class)
Number of Features	4 continuous numerical dimensions
Number of Classes	3 (Iris-setosa, Iris-versicolor, Iris-virginica)
Missing Values	0 missing records
Train-Test Split	N/A (Exploratory Analysis Phase)

The dataset attributes include:

- **SepalLengthCm**: Sepal length in centimeters.
- **SepalWidthCm**: Sepal width in centimeters.
- **PetalLengthCm**: Petal length in centimeters.

- **PetalWidthCm**: Petal width in centimeters.
- **Species**: Iris species category (Target).

4 Exploratory Data Analysis

The automated EDA module generated 17 visualizations saved in `figures_iris/`.

4.1 Histograms of Numerical Features

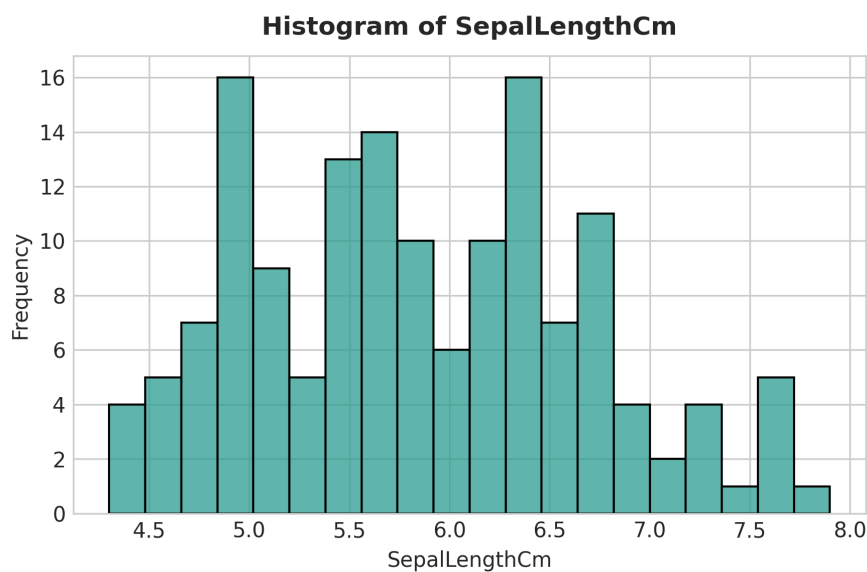


Figure 1: Histogram of Sepal Length

Inference: Sepal length ranges from 4.3 cm to 7.9 cm, forming a unimodal distribution centered near 5.8 cm.

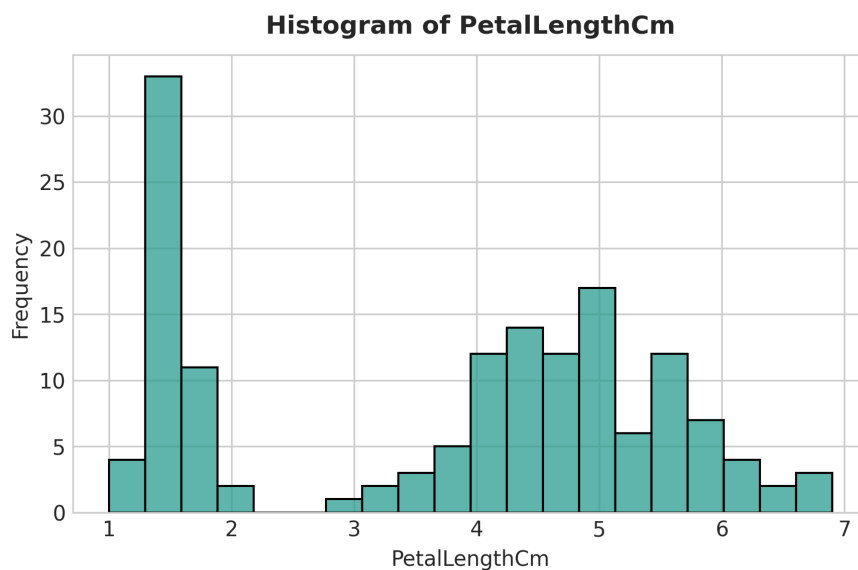


Figure 2: Histogram of Petal Length

Inference: Petal length shows a distinct bimodal distribution with a clear density gap between 2.0 cm and 3.0 cm, isolating *Iris-setosa* from *Versicolor* and *Virginica*.

4.2 Box Plots of Numerical Features

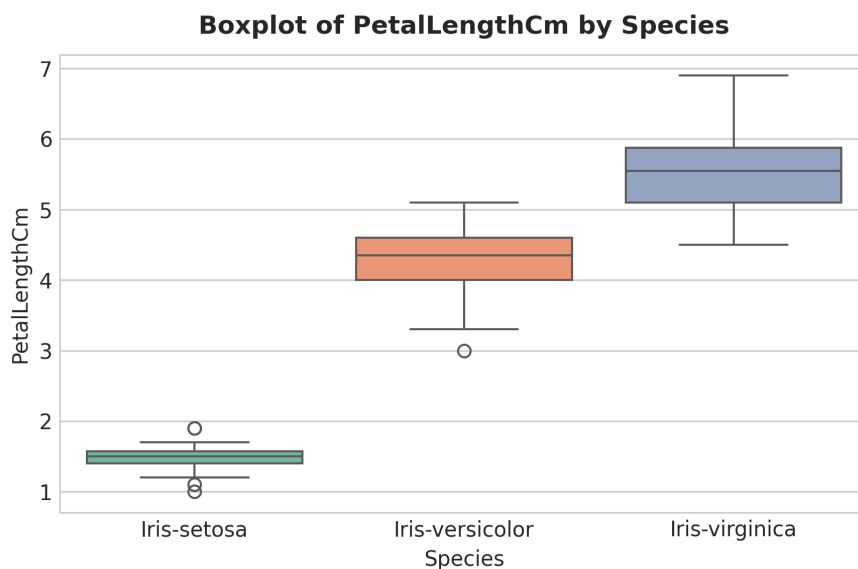


Figure 3: Box Plot of Petal Length by Species

Inference: *Iris-setosa* features significantly shorter petals (1.0–1.9 cm) with non-overlapping interquartile ranges relative to *Versicolor* (3.0–5.1 cm) and *Virginica* (4.5–6.9 cm).

4.3 Kernel Density Estimation (KDE) Plots

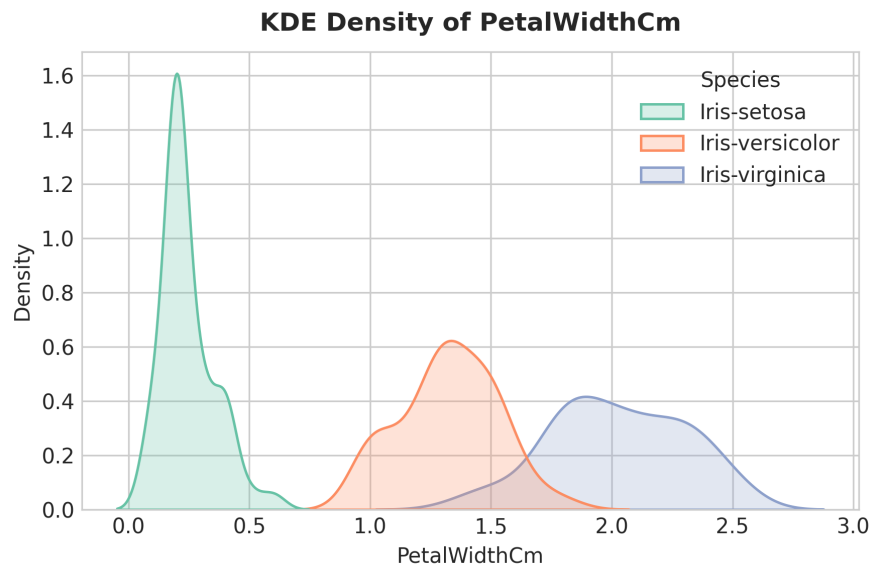


Figure 4: KDE Density of Petal Width by Species

Inference: Density estimation highlights complete linear separability of Setosa along petal width, while Versicolor and Virginica display partial overlap in the 1.4–1.8 cm interval.

4.4 Count Plots of Categorical Features

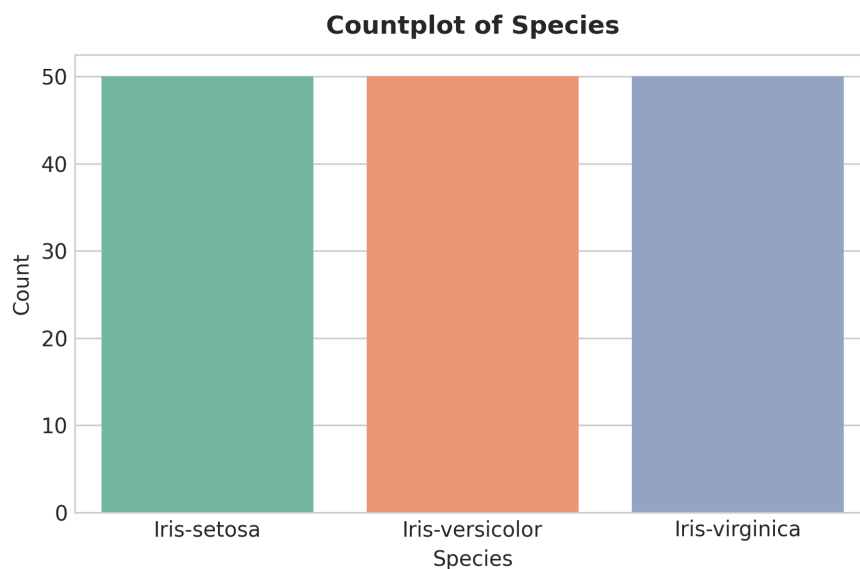


Figure 5: Count Plot of Iris Species

Inference: Displays a perfectly balanced dataset containing exactly 50 samples for each species class.

4.5 Correlation Heatmap of Numerical Features

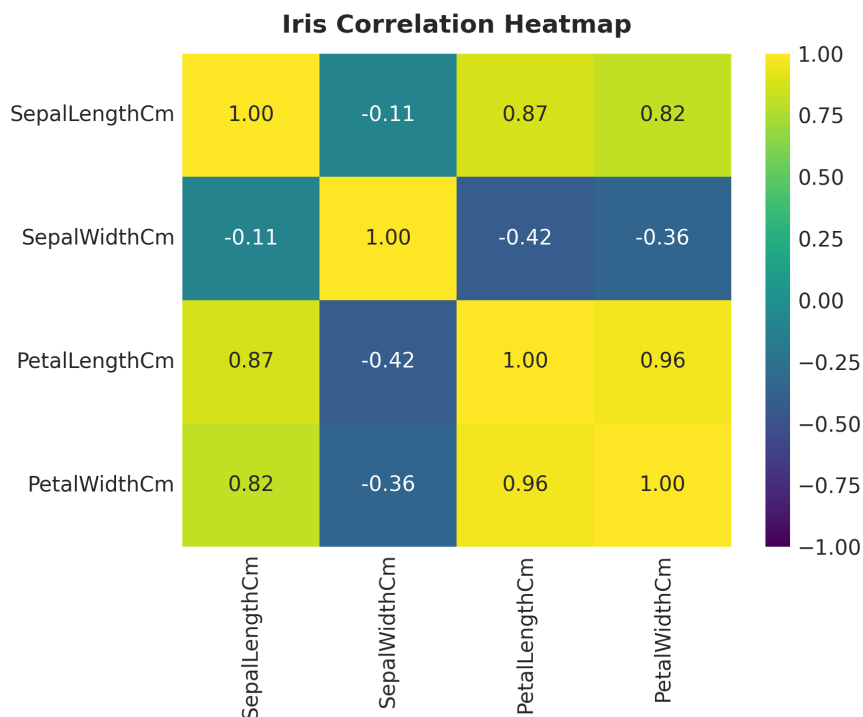


Figure 6: Correlation Heatmap of Iris Features

Inference: Petal length and petal width display a high positive correlation ($r = 0.96$). Sepal length also correlates strongly with petal dimensions ($r > 0.81$).

4.6 Pairwise Scatter Matrix (Pair Plot)



Figure 7: Pair Plot of Iris Features by Species

Inference: The pairplot grid visually confirms that Setosa forms a linearly isolated cluster in every bivariate projection, whereas Versicolor and Virginica require non-linear or multi-feature boundaries.

4.7 Bivariate Scatter Plots

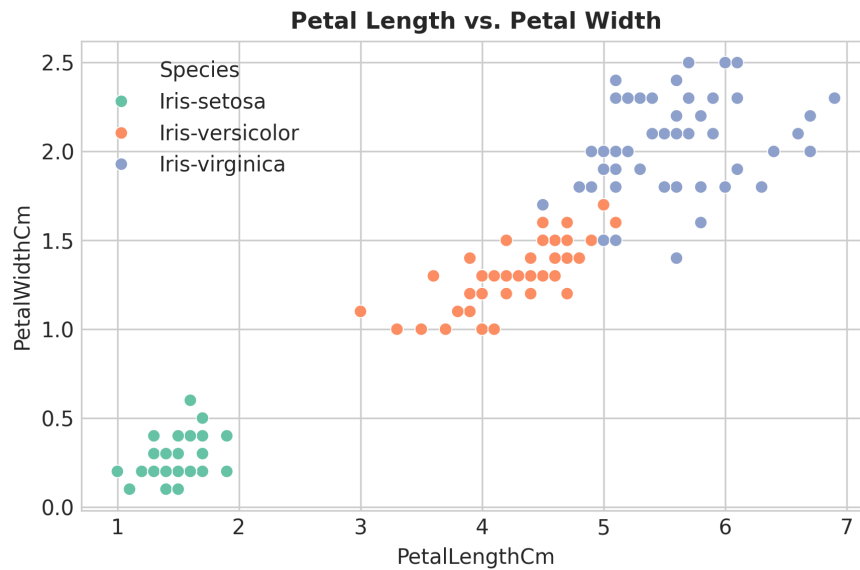


Figure 8: Scatter Plot: Petal Length vs. Petal Width

Inference: Displays strong linear clustering per species, confirming petal dimensions as the dominant discriminative feature pair.

5 Data Preprocessing

The Iris dataset contains zero missing values and zero duplicate samples. Feature standardization using Z-score scaling ($z = \frac{x - \mu}{\sigma}$) is recommended prior to training distance-sensitive algorithms like K-Nearest Neighbors (KNN) or Support Vector Machines (SVM).

6 Machine Learning Algorithms

- **Suitable Models:** Decision Trees, Random Forests, Support Vector Machines (RBF/Linear kernel), K-Nearest Neighbors (KNN).
- **Feature Selection:** ANOVA F-test via `SelectKBest` or Principal Component Analysis (PCA) for dimensionality reduction.

7 Experimental Setup

Executed on Python 3.10+ using Pandas, NumPy, Matplotlib, and Seaborn on Windows 11.

8 Hyperparameter Selection

Density kernel bandwidths and pairwise bin sizing were determined dynamically using Seaborn defaults.

9 Results

All 17 visualization outputs were successfully generated and stored in `figures_iris/`.

10 Result Visualization

The visualizations demonstrate that petal dimensions (`PetalLengthCm` and `PetalWidthCm`) provide clean multi-class discrimination boundaries.

11 Discussion

The Iris dataset provides a foundational benchmark for multiclass linear vs. non-linear decision boundaries.

- **Strengths:** Small dimension, balanced classes, zero null values.
- **Insights:** Setosa requires only simple threshold rules, whereas Virginica and Versicolor require soft-margin classifiers.

12 Conclusion

The automated EDA framework effectively characterized the Iris dataset, establishing that petal length and width serve as optimal predictive attributes for species classification.

13 References

- [1] Fisher, R. A., “The use of multiple measurements in taxonomic problems,” *Annals of Eugenics*, 1936.
- [2] UCI Machine Learning Repository, “Iris Dataset,” <https://archive.ics.uci.edu/ml/datasets/iris>.